Electrical Storm

Section I: Scenario Demographics

Scenario Title:	Electrical Storm
Date of Development:	27/12/2015 (DD/MM/YYYY)
Target Learning Group:	☒ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	P. Dieckmann & M. Rall
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to the management of refractory VF (electrical storm)
CRM Objectives:	Effectively lead a code with multiple team members
Medical Objectives:	1. Manage cardiac arrest with standard ACLS principles 2. Identify and appropriately manage electrical storm 3. Initiate appropriate post-arrest care

Case Summary: Brief Summary of Case Progression and Major Events
A 55 year-old male is brought to the emergency department with absent vital signs. He collapsed at his office after complaining of feeling unwell. CPR was started by a colleague and continued by EMS. He received 3 shocks by an AED. His downtime is approximately 10 minutes. The team is expected to perform routine ACLS care. When the patient remains in VF despite ACLS management, the team will need to consider specific therapies, such as iv beta blockade or dual sequential shock, in order to abort the electrical storm.

Facilitators Required to Run Session
Instructors (faculty or senior resident): 1-2 (one to observe, one to run mannequin if no sim tech)
Confederate nurse: 1 (to assist at bedside)
Sim techs: 1

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
Sharif, S, Owen JJ. ß-Blockers for refractory ventricular fibrillation in cardiac arrest. *CanadiEM* (2017). < https://canadiem.org/beta-blockers-in-cardiac-arrest/>

Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates	Brief Description of Role
Nurse	To assist at bedside, cue team to patient's status PRN

B. Required Monitors		
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads **X 2**	☒ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

C. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☒ Intraosseous Set-up	☒ LMA	☐ Other:

D. Moulage
None

E. Approximate Timing					
Set-Up:	5 min	Scenario:	15 min	Debriefing:	20 min

Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case
Arrest arriving in 1 minute. Doctor to resuscitation room STAT. Paramedic report: *"This is a 55 year old male we picked up at an office tower down the street. Apparently he was complaining of feeling unwell all morning and then collapsed at lunch. A colleague started CPR and we were called. The AED delivered 3 shocks. His colleagues say he's healthy and they're unsure about meds or allergies. His boss called his wife and she's on her way."* CPR is ongoing.

B. Patient Profile and History			
Patient Name: Michael Johnson	Age: 55	Weight: 70 kg	
Gender: ☒ M ☐ F	Code Status: Full Code		
Chief Complaint: Cardiac Arrest			
History of Presenting Illness: Feeling unwell and collapse at lunch			
Past Medical History:	Unknown	Medications:	Unknown
Allergies: Unknown			
Social History: Works at an office; all else unknown			
Family History: Unknown			

Review of Systems:				
	CNS:	GCS 3		
	HEENT:	Nil		
	CVS:	VF on monitor; Patient is pulseless		
	RESP:	O2 sat 78% with CPR		
	GI:	Nil		
	GU:	Nil		
	MSK:	Nil	INT:	Nil

C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 0/min	BP: 0	RR: 0/min	O_2SAT: 78%
Rhythm: VF	T: 35.6°C	Glucose: 6.2 mmol/L	GCS: 3 (E1 V1 M1)
General Status: Obtunded			

CNS:	**Pupils dilated at 7mm bilaterally; not reactive to light**		
HEENT:	Nil acute		
CVS:	Inaudible heart sounds		
RESP:	Good air entry with bagging		
ABDO:	Soft, Non-distended		
GU:	Nil acute		
MSK:	Not moving any limbs	SKIN:	**Pale, mottled**

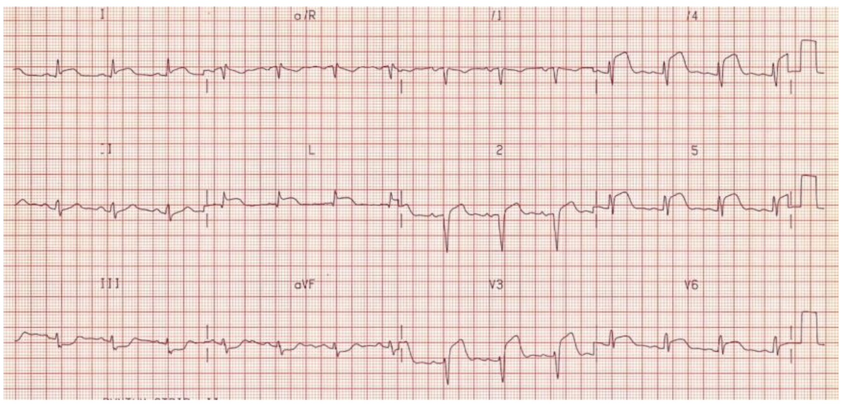
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: VF HR: 0/min BP: 0/0 RR: 0/min O_2SAT: 78% with CPR T: 35.6°C $ETCO_2$: 20	Obtunded, non-responsive	<u>Learner Actions</u> -☐ High quality CPR -☐ Monitors & Defibrillator -☐ IV access -☐ BVM -☐ Defibrillate VF as per ACLS -☐ ± Bedside Echo -☐ Give 1mg Epi q3-5 minutes -☐ Amiodarone as per ACLS -☐ ± Bicarbonate -☐ Give IV fluid bolus -☐ Labs including extended lytes, VBG, lactate, troponin -☐ ± Consider intubation or LMA	<u>Modifiers</u> *Changes to patient condition based on learner action* - Patient remains in VF regardless of defibrillation attempts or any other treatments <u>Triggers</u> *For progression to next state* - If defibrillated x 4, amiodarone given x 2 → **2. Refractory VF**
2. Refractory VF Unchanged	If learners not yet considering tailored refractory VF treatments, **RN to cue at state onset** "isn't there a different medication you can give? Or some weird way of shocking?"	<u>Learner Actions</u> -☐ Shock with Paddles -☐ Double-Sequential Defibrillation -☐ ß-Blocker (Esmolol) -☐ ECMO (not available) -☐ ± Lidocaine -☐ ± Magnesium (1-2 grams) -☐ ± Bicarbonate (1 amp) -☐ ± Calcium Chloride (1 amp)	<u>Modifiers</u> <u>Triggers</u> - Esmolol given, shock with paddles or double sequential defib → **3. ROSC** **-** If none of above done after 6 defibs → **4. Asystole**
3. ROSC HR → 110 BP → 100/60 RR → 12 (intubated) O_2SAT → 90% $ETCO_2$ → 38	Patient's colour improves	<u>Learner Actions</u> -☐ EKG (12 and 15 lead) -☐ Call Cath lab/ICU/CCU -☐ Targeted temperature management -☐ Optimize hemodynamics -☐ Intubate if not yet done -☐ Post-Intubation CXR -☐ Initiate sedation	<u>Modifiers</u> - EKG → STEMI <u>Triggers</u> - All actions complete or 15 min→ **(End Case – "cath lab is ready"**)
4. Asystole Rhythm: Asystole HR → 0 $ETCO_2$ → 5	Non-responsive	<u>Learner Actions</u> -☐ Epinephrine 1mg q3-5min -☐ ± Bicarbonate -☐ ± Calcium chloride -☐ ± Consider ending resuscitation	**END CASE PRN**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

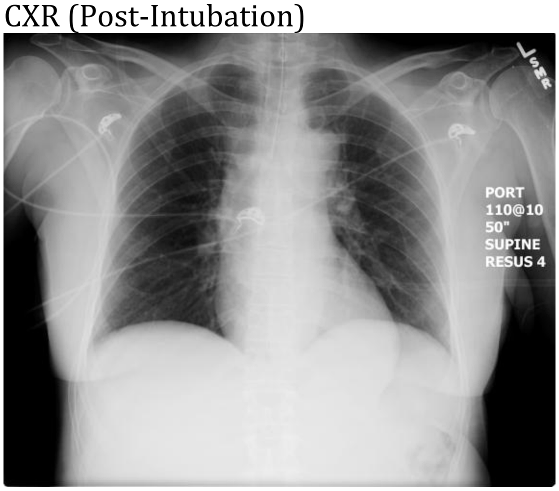
Laboratory Results
No labs back

Images (ECGs, CXRs, etc.)
EKG: Anterior STEMI



Source: https://lifeinthefastlane.com/wp-content/uploads/2012/01/anterolateral.jpg

CXR (Post-Intubation)



Source: https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg

Ultrasound Video Files (if applicable)
Ventricular Fibrillation on Echo

Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing

1. What is the definition of refractory ventricular fibrillation?
2. How is refractory ventricular fibrillation managed?
3. How do you perform double-sequential defibrillation?
4. What are the indications and contraindications to thrombolysis?
5. How did you feel the communication was with the team leader and the rest of the team?
6. How did your team handle this complex case? Were you all aware what the plan was?

Key Moments
Recognition of refractory VF
Initiating adjunctive therapies (i.e. dual-sequential defibrillation, ß-Blockers)
Post-ROSC → Initiate targeted temperature management, Order EKG and Call Cath Lab